

Convexity & Concavity

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UAB-IDEA

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Section 1

Convex sets

Convex combination: definition

Let $m \geq 1$ and let $x_1, \dots, x_m \in \mathbb{R}^n$.

Convex combination. A point $z \in \mathbb{R}^n$ is a convex combination of $x_1, \dots, x_m$ if there are weights $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ such that

$$z = \sum_{i=1}^m \lambda_i x_i, \quad \lambda_i \geq 0 \text{ for every } i, \quad \sum_{i=1}^m \lambda_i = 1.$$

Video: Dr. Will Wood, *Convex sets: introduction, definition and examples*.

NONNEGATIVE WEIGHTS

Every point enters with a weight that is at least zero.

WEIGHTS SUM TO ONE

The coefficients form a weighted average of the points.

Linear, affine, conic, and convex combinations

Start from the coefficients, then impose restrictions

For $x_1, \dots, x_m \in \mathbb{R}^n$, start from $z = \sum_{i=1}^m \lambda_i x_i$ with $\lambda_i \in \mathbb{R}$, then classify the combination by the coefficient conditions.

Combination	Coefficient condition	Sum condition
Linear	$\lambda_i \in \mathbb{R}$	none
Affine	$\lambda_i \in \mathbb{R}$	$\sum_{i=1}^m \lambda_i = 1$
Conic	$\lambda_i \geq 0$ for every i	none
Convex	$\lambda_i \geq 0$ for every i	$\sum_{i=1}^m \lambda_i = 1$

A convex combination satisfies $\sum_i \lambda_i = 1$, as an affine combination does, and $\lambda_i \geq 0$, as a conic combination does. For two distinct points x, y , $\lambda \in \mathbb{R}$ traces the full line through them, while $\lambda \in [0, 1]$ traces the joining segment.

A convex combination in $\mathbb{R}^2$

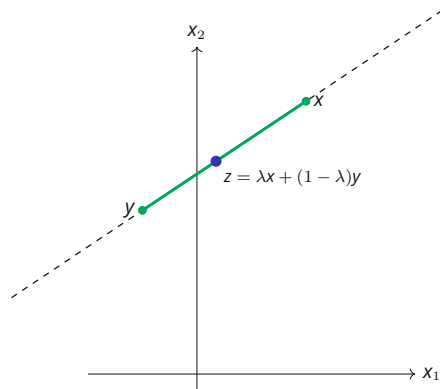


Figure 1 · A segment in $\mathbb{R}^2$.

A convex combination of x and y is any point on the segment joining them.

Convex combinations

POINTS

$$x_1 = (0, 0), \quad x_2 = (2, 0), \quad x_3 = (0, 2).$$

WEIGHTS

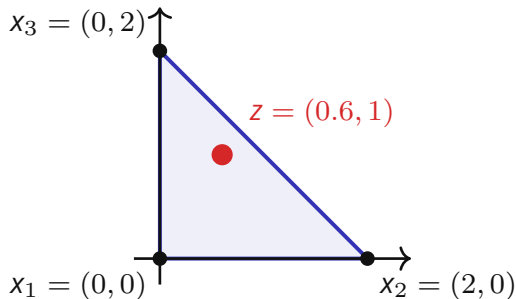
$$(\lambda_1, \lambda_2, \lambda_3) = (0.2, 0.3, 0.5).$$

All three weights are nonnegative, and

$$\lambda_1 + \lambda_2 + \lambda_3 = 1.$$

CALCULATION

$$\begin{aligned} z &= 0.2x_1 + 0.3x_2 + 0.5x_3 \\ &= (0.6, 1). \end{aligned}$$



The shaded triangle is the set of all convex combinations of x_1, x_2, x_3 . The red circle marks a particular point z from them.

Definition

Convex set. $S \subseteq \mathbb{R}^n$ is convex if

$$\forall x, y \in S, \forall \lambda \in [0, 1] : \quad \lambda x + (1 - \lambda)y \in S.$$

- ◆ Equivalently, S contains the entire segment between **any** two of its points.
- ◆ Visually: every line segment joining two points of S stays inside S .

Extreme points cannot be decomposed inside the set

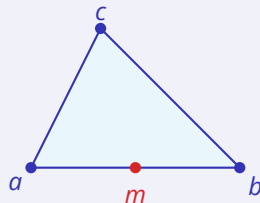
Extreme points

A point $x \in C$ is **extreme** if

$$x = \lambda y + (1 - \lambda)z, \quad y, z \in C, \quad \lambda \in (0, 1)$$

implies $y = z = x$.

For noncollinear a, b, c , the three vertices of $C = \text{conv}\{a, b, c\}$ are extreme. The midpoint $m = \frac{1}{2}a + \frac{1}{2}b$ is not.



Convex hull: generating a convex set from a non-convex set

Convex hull of a set

For $A \subseteq \mathbb{R}^n$,

$$\text{conv}(A) = \left\{ \sum_{i=1}^m \lambda_i x_i : m \geq 1, x_i \in A, \lambda_i \geq 0, \sum_{i=1}^m \lambda_i = 1 \right\}.$$

Equivalently, $\text{conv}(A)$ is the smallest convex set containing A .

Further reading: [Boyd and Vandenberghe, convex-set slides](#).

Budget set

Convexity of a budget set

Let

$$B = \{x \in \mathbb{R}_+^2 : p_1 x_1 + p_2 x_2 \leq m\}, \quad p_1, p_2, m > 0.$$

Also,

Take any $x, y \in B$ and $\lambda \in [0, 1]$. Since $x, y \geq 0$,

$$\lambda x + (1 - \lambda)y \geq 0.$$

$$\begin{aligned} p \cdot (\lambda x + (1 - \lambda)y) &= \lambda p \cdot x + (1 - \lambda)p \cdot y \\ &\leq \lambda m + (1 - \lambda)m \\ &= m. \end{aligned}$$

Therefore $\lambda x + (1 - \lambda)y \in B$, so B is convex.

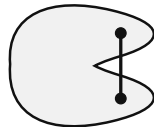
Convex & non-convex sets



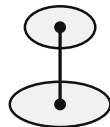
(a) convex



(b) convex



(c) not convex



(d) not convex

Basic properties

- ◆ **Arbitrary intersections** of convex sets are convex.
- ◆ **Linear combinations of convex sets are convex:** if S, T are convex and $\alpha, \beta \in \mathbb{R}$, then $\alpha S + \beta T = \{\alpha x + \beta y : x \in S, y \in T\}$ is convex.
- ◆ **Unions** of convex sets **need not** be convex.

Intersections

Let

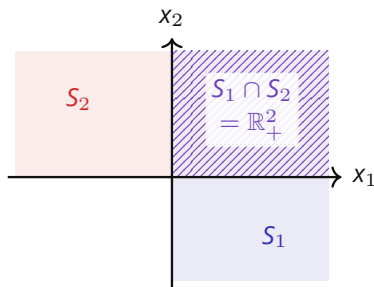
$$S_1 = \{x \in \mathbb{R}^2 : x_1 \geq 0\},$$

$$S_2 = \{x \in \mathbb{R}^2 : x_2 \geq 0\}.$$

A point belongs to $S_1 \cap S_2$ precisely when both coordinates are nonnegative. Therefore

$$S_1 \cap S_2 = \mathbb{R}_+^2,$$

which is convex.



The hatched first quadrant marks the points satisfying both inequalities.

Video: [Stanford EE364A, Lecture 2: convex sets and operations that preserve convexity.](#)

Intersections of convex sets

Let $\{S_i : i \in I\}$ be a collection of convex subsets of $\mathbb{R}^n$. Prove directly from the definition that

$$\bigcap_{i \in I} S_i$$

is convex.

Intersections of convex sets

If $x, y \in \bigcap_{i \in I} S_i$ and $t \in [0, 1]$, then $x, y \in S_i$ for every $i \in I$. Convexity of each S_i gives

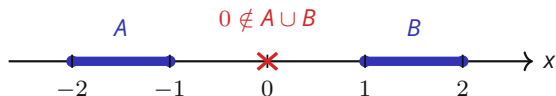
$$tx + (1 - t)y \in S_i \quad \text{for every } i \in I.$$

Hence

$$tx + (1 - t)y \in \bigcap_{i \in I} S_i,$$

so the intersection is convex.

Unions need not be convex



Take two convex intervals:

$$A = [-2, -1], \quad B = [1, 2].$$

Their union contains -1 and 1 , but it does not contain their midpoint:

$$-1, 1 \in A \cup B, \quad \frac{-1 + 1}{2} = 0 \notin A \cup B.$$

Hence $A \cup B$ is not convex.

A resource set and two subsets

Let

$$T = \{(x, y) \in \mathbb{R}^2 : 2x + y \leq 6\}, \quad C = \{(0, 0), (3, 0)\} \subset T.$$

- (a) Prove directly from the definition that T is convex.
- (b) Determine whether $T \cap C$ and $T \cup C$ are convex. Justify both answers.
- (c) Give a convex subset $D \subset T$, with at least two points, for which both $T \cap D$ and $T \cup D$ are convex.

A resource set and two subsets

If $u, v \in T$ and $t \in [0, 1]$, then

$$2(tu_1 + (1-t)v_1) + tu_2 + (1-t)v_2 = t(2u_1 + u_2) + (1-t)(2v_1 + v_2) \leq 6.$$

Thus $tu + (1-t)v \in T$, so T is convex. Since $C \subset T$,

$$T \cap C = C, \quad T \cup C = T.$$

The set C is not convex because $(3/2, 0) \notin C$, although it is the midpoint of $(0, 0)$ and $(3, 0)$. Hence $T \cap C$ is not convex, while $T \cup C$ is convex. One valid choice in part (c) is $D = \{(x, 0) : 0 \leq x \leq 3\}$: then $T \cap D = D$ and $T \cup D = T$.

Production sets and the profit function

Production plans and profit

A production set $Y \subseteq \mathbb{R}^n$ contains feasible netput vectors, with positive coordinates for outputs and negative coordinates for inputs. If Y is convex, mixtures of feasible production plans remain feasible.

$$\pi(p) = \sup_{y \in Y} p \cdot y, \quad p \in \mathbb{R}_+^n.$$

As a supremum of linear functions of p , π is convex wherever it is finite. Convexity of Y instead concerns feasible mixtures.

Profit function

Support function

Let

$$Y = \text{conv} \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \end{bmatrix} \right\}, \quad p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The linear payoff at the three vertices is

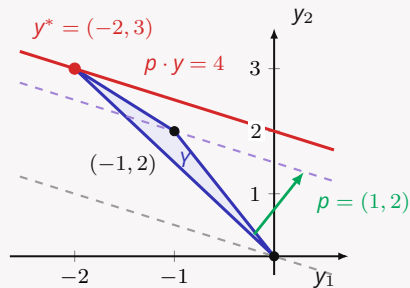
$$p \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0, \quad p \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = 3, \quad p \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix} = 4.$$

If $y = \sum_i \lambda_i y_i \in Y$, then linearity gives

$$p \cdot y = \sum_i \lambda_i (p \cdot y_i) \leq \max_i p \cdot y_i = 4.$$

Hence

$$\pi(p) = \sup_{y \in Y} p \cdot y = 4, \quad y^* = \begin{bmatrix} -2 \\ 3 \end{bmatrix}.$$



The level line $p \cdot y = 4$ supports Y at y^* .

Disjoint compact and closed convex sets can be strictly separated

Separating hyperplanes

If $A \subset \mathbb{R}^n$ is nonempty, compact, and convex, and $B \subset \mathbb{R}^n$ is nonempty, closed, and convex, with $A \cap B = \emptyset$, then there exist $p \neq 0$ and $\alpha < \beta$ such that

$$p \cdot a \leq \alpha < \beta \leq p \cdot b \quad \forall a \in A, \forall b \in B.$$

Every level set $\{x : p \cdot x = t\}$ with $t \in (\alpha, \beta)$ is a separating hyperplane. The normal p gives the direction in which the two sets are ordered.

For the discs centered at $(-2, 0)$ and $(2, 0)$, both of radius one, $p = (1, 0)$ gives

$$\sup_{a \in A} p \cdot a = -1 < 1 = \inf_{b \in B} p \cdot b,$$

so $p \cdot x = 0$ separates them.

Video: [Constantine Caramanis, Separating hyperplanes.](#)

Every boundary point of a closed convex set has a support

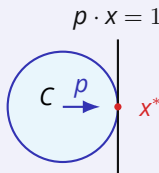
Supporting hyperplanes

Let $C \subset \mathbb{R}^n$ be nonempty, closed, and convex, and let $x^* \in \partial C$. There exists $p \neq 0$ such that

$$p \cdot x \leq p \cdot x^* \quad \forall x \in C.$$

The hyperplane $p \cdot x = p \cdot x^*$ supports C at x^* : it contains the boundary point and keeps the entire convex set in one closed half-space.

For the unit disc, $x^* = (1, 0)$ and $p = (1, 0)$ give the supporting line $p \cdot x = 1$.



Section 2

Convex & concave functions

Definitions via convex combinations

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$. Then the function f is

Convex. $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in S$ and $\lambda \in [0, 1]$.

Concave. $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in S$ and $\lambda \in [0, 1]$.

Strict convexity or concavity replaces the weak inequality by a strict one when $x \neq y$ and $\lambda \in (0, 1)$.

Here, convex and concave functions are defined only on convex domains. (Things are different in discrete convex analysis, which develops alternative notions of convexity for discrete domains.)

Video: [Arizona Math Camp, Concave and convex functions.](#)

Chord above, chord below

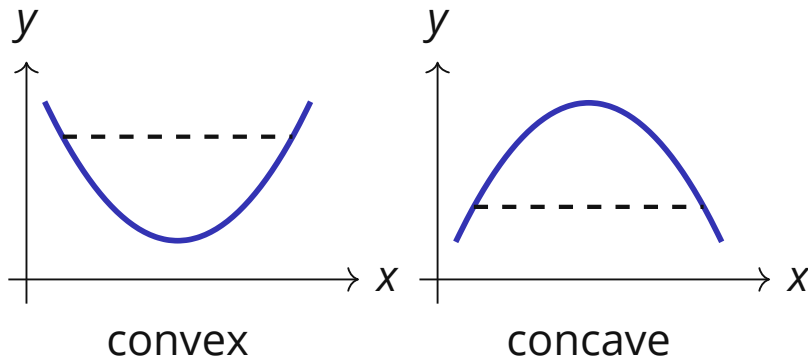


Figure 3 · Secants of two graphs.

The dashed secant lies above the convex graph and below the concave graph.

Degree-one homogeneity rules out strict concavity

Homogeneity is incompatible with strict concavity

Let D be a convex cone containing a nonzero x , and let $f : D \rightarrow \mathbb{R}$ be homogeneous of degree one. Choose $y = 2x$. Then $x \neq y$, while

$$\begin{aligned} f\left(\frac{1}{2}x + \frac{1}{2}y\right) &= f\left(\frac{3}{2}x\right) = \frac{3}{2}f(x), \\ \frac{1}{2}f(x) + \frac{1}{2}f(y) &= \frac{1}{2}f(x) + \frac{1}{2}f(2x) = \frac{3}{2}f(x). \end{aligned}$$

The two sides of the strict-concavity inequality are therefore equal for two distinct points, so strict concavity fails. A degree-one homogeneous function on a nontrivial convex cone is therefore never strictly concave. The same calculation rules out strict convexity.

Source: [Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., §2.3, Problem 6\(c\)](#); the $x, 2x$ argument follows the Student's Manual solution to that problem.

Convexity from the definition

Let $f(x) = x^2$. To verify convexity, compute

$$\begin{aligned} & \lambda x^2 + (1 - \lambda)y^2 - (\lambda x + (1 - \lambda)y)^2 \\ &= \lambda(1 - \lambda)x^2 - 2\lambda(1 - \lambda)xy + \lambda(1 - \lambda)y^2 \\ &= \lambda(1 - \lambda)(x - y)^2 \geq 0. \end{aligned}$$

Therefore

$$(\lambda x + (1 - \lambda)y)^2 \leq \lambda x^2 + (1 - \lambda)y^2.$$

If $x \neq y$ and $\lambda \in (0, 1)$, the inequality is strict. Hence x^2 is strictly convex.

Affine functions: boundary case

Let $f(x) = a^\top x + b$. Then

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &= a^\top (\lambda x + (1 - \lambda)y) + b \\ &= \lambda a^\top x + (1 - \lambda)a^\top y + b \\ &= \lambda f(x) + (1 - \lambda)f(y). \end{aligned}$$

The convexity inequality and the concavity inequality both hold with equality.

Conversely, if $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is both convex and concave, the two inequalities force equality for every x, y and $\lambda \in [0, 1]$. For $g = f - f(0)$, this equality gives $g(x + y) = g(x) + g(y)$ and $g(tx) = tg(x)$ for every $t \in \mathbb{R}$; hence $g(x) = a^\top x$ and f is affine.

Thus **a function is affine if and only if it is both convex and concave.**

In particular, when $b = 0$, $f(x) = a^\top x$ is linear; hence every linear function is both convex and concave.

Jensen's inequality

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$ convex. For $x_1, \dots, x_m \in S$ and $\lambda_i \geq 0$ with $\sum_i \lambda_i = 1$,

$$f\left(\sum_{i=1}^m \lambda_i x_i\right) \leq \sum_{i=1}^m \lambda_i f(x_i).$$

For a concave function, the inequality is reversed.

Video: [MIT OpenCourseWare, Jensen's inequality](#).

Jensen for random vectors

From finite mixtures to expectations

If X takes values in S , the expectations below exist, and $\mathbb{E}[X] \in S$, then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

for convex f , with the reverse inequality for concave f .

Further reading: [Boyd and Vandenberghe, Jensen's inequality](#).

Video: [Stanford EE364A, Lecture 3: convex functions, Hessian tests, epigraphs and Jensen's inequality](#).

Jensen inequality

Jensen's inequality for a two-point distribution

Let $f(x) = x^2$ and let X take values 1 and 3 with probability $1/2$ each.

FUNCTION AT THE MEAN

$$\mathbb{E}[X] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 3 = 2,$$

$$f(\mathbb{E}[X]) = f(2) = 4.$$

MEAN OF THE FUNCTION

$$\begin{aligned}\mathbb{E}[f(X)] &= \frac{1}{2}f(1) + \frac{1}{2}f(3) \\ &= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 9 \\ &= 5.\end{aligned}$$

$$\text{Hence } f(\mathbb{E}[X]) = 4 < 5 = \mathbb{E}[f(X)].$$

Epigraph & hypograph

Epigraph and hypograph tests

Let $S \subseteq \mathbb{R}^n$ be convex and let $f: S \rightarrow \mathbb{R}$.

Epigraph.

$$\text{epi } f = \{(x, t) \in S \times \mathbb{R} : t \geq f(x)\}.$$

Hypograph.

$$\text{hypo } f = \{(x, t) \in S \times \mathbb{R} : t \leq f(x)\}.$$

$$f \text{ convex} \Leftrightarrow \text{epi } f \text{ convex}, \quad f \text{ concave} \Leftrightarrow \text{hypo } f \text{ convex}.$$

Function convexity is convexity of the epigraph, a set one dimension higher. Also, f is convex iff $-f$ is concave, so the epigraph and hypograph statements are mirror images.

Further reading: [Boyd and Vandenberghe, convex-function slides](#).

Convex functions, Jensen, and epigraphs

- (a) Use the definition to show that $f(x) = |x|$ is convex on $\mathbb{R}$.
- (b) Use Jensen's inequality to prove

$$\left(\frac{x+y}{2}\right)^2 \leq \frac{x^2+y^2}{2}.$$

When does equality hold?

- (c) Describe $\text{epi } f$ for $f(x) = x^2$, and use the epigraph test to explain why it is convex.

Convex functions, Jensen, and epigraphs

The triangle inequality gives

$$|tx + (1-t)y| \leq t|x| + (1-t)|y|, \quad 0 \leq t \leq 1.$$

For $q(z) = z^2$, Jensen gives the requested inequality; equivalently,

$$\frac{x^2 + y^2}{2} - \left(\frac{x+y}{2}\right)^2 = \frac{(x-y)^2}{4} \geq 0,$$

with equality exactly when $x = y$. Finally,

$$\text{epi } f = \{(x, s) \in \mathbb{R}^2 : s \geq x^2\}.$$

If $s_i \geq x_i^2$, convexity of x^2 implies $ts_1 + (1-t)s_2 \geq (tx_1 + (1-t)x_2)^2$, so the epigraph is convex.

Randomization and cost

Randomization produces convex costs

Let $C(\mu)$ be the minimum cost of producing distribution μ , assume the minima are attained, and suppose randomization across procedures is feasible.

Randomizing between cost-minimizing procedures for μ and ν with probabilities α and $1 - \alpha$ produces $\alpha\mu + (1 - \alpha)\nu$ at expected cost $\alpha C(\mu) + (1 - \alpha)C(\nu)$. Hence:

$$C(\alpha\mu + (1 - \alpha)\nu) \leq \alpha C(\mu) + (1 - \alpha)C(\nu).$$

Randomization produces the convexity inequality for C by mixing procedures.

Finite randomization convexifies feasible outcomes and costs

General convexification

If deterministic procedures generate outcomes in A , finite randomization generates every point in $\text{conv}(A)$. For a bounded-below deterministic cost $c : A \rightarrow \mathbb{R}$, the infimum expected cost of generating $x \in \text{conv}(A)$ is

$$c^{\text{cvx}}(x) = \inf \left\{ \sum_{i=1}^m \lambda_i c(x_i) : m \geq 1, \sum_{i=1}^m \lambda_i x_i = x, x_i \in A, \lambda_i \geq 0, \sum_{i=1}^m \lambda_i = 1 \right\}.$$

This convex envelope is the largest convex function on $\text{conv}(A)$ that does not exceed c on A .

The concave envelope is the smallest concave majorant

Concave envelopes

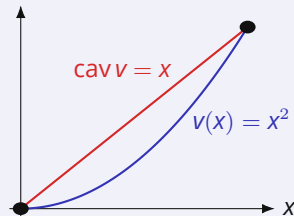
Let $D \subseteq \mathbb{R}^n$ be convex and let $v : D \rightarrow \mathbb{R}$ be bounded above. Its smallest concave majorant is

$$\text{cav } v(x) = \sup_{\substack{m \geq 1, x_i \in D, \lambda_i \geq 0 \\ \sum_i \lambda_i = 1, \sum_i \lambda_i x_i = x}} \sum_i \lambda_i v(x_i).$$

On $D = [0, 1]$, let $v(x) = x^2$. The affine function $h(x) = x$ majorizes v . Every concave majorant satisfies

$$h(x) \geq (1-x)h(0) + xh(1) \geq x,$$

so $\text{cav } v(x) = x$.



Section 3

First-order characterizations

First-order characterization in $\mathbb{R}^n$

Let $S \subseteq \mathbb{R}^n$ be **open and convex**, and let $f : S \rightarrow \mathbb{R}$ be differentiable. Then

Convex iff.

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in S.$$

Concave iff.

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in S.$$

For $n = 1$, these are the familiar tangent-line inequalities. In particular, for convex f , $\nabla f(x^*) = 0 \Rightarrow f(y) \geq f(x^*)$ for every $y \in S$; the concave case reverses the inequality.

Video: [Mike, the Mathematician, First Order Conditions for Convexity.](#)

First-order characterization: supporting tangent planes

For a convex function

$$f(u, v) = \frac{1}{2}(u^2 + v^2), \quad \bar{x} = (1, 0),$$

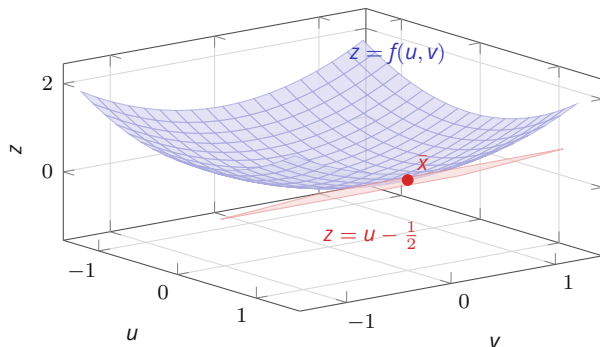
we have $f(\bar{x}) = \frac{1}{2}$ and $\nabla f(\bar{x}) = (1, 0)^\top$. The tangent plane is

$$L(u, v) = f(\bar{x}) + \nabla f(\bar{x})^\top \begin{bmatrix} u - 1 \\ v \end{bmatrix} = u - \frac{1}{2}.$$

Moreover,

$$f(u, v) - L(u, v) = \frac{1}{2}((u - 1)^2 + v^2) \geq 0.$$

Thus, for convex f , every tangent plane supports the graph from below. For concave f , every tangent plane supports the graph from above.



The graph of a convex function f lies above its tangent plane at $\bar{x}$.

Tangent test

CONVEX FUNCTION For $f(x) = x^2$ at $x_0 = 1$, the tangent line is

$$f(1) + f'(1)(x - 1) = 1 + 2(x - 1) = 2x - 1.$$

The graph stays above it:

$$x^2 - (2x - 1) = (x - 1)^2 \geq 0.$$

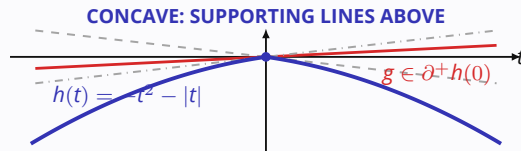
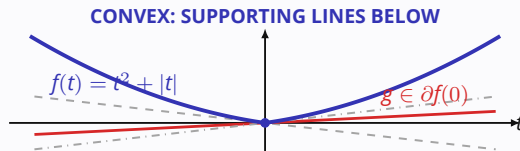
CONCAVE FUNCTION For $g(x) = \ln x$ at $x_0 = 1$, the tangent line is

$$g(1) + g'(1)(x - 1) = x - 1.$$

Concavity gives the global upper bound

$$\ln x \leq x - 1, \quad x > 0.$$

Subgradients & supergradients



First-order bounds without differentiability

Let $S \subseteq \mathbb{R}^n$ be open and convex, $x \in S$, and $f : S \rightarrow \mathbb{R}$.

Subdifferential for convex f .

$$\partial f(x) = \{g \in \mathbb{R}^n : f(y) \geq f(x) + g^\top (y - x), \forall y \in S\}.$$

Superdifferential for concave f .

$$\partial^+ f(x) = \{g \in \mathbb{R}^n : f(y) \leq f(x) + g^\top (y - x), \forall y \in S\}.$$

If f is differentiable at x . $\partial f(x) = \{\nabla f(x)\}$ (convex), $\partial^+ f(x) = \{\nabla f(x)\}$ (concave).

Video: Kiet, *Understanding Subgradients Using Examples*.

Subgradients of the absolute value function

Let $f(x) = |x|$.

- (a) Find $\partial f(0)$.
- (b) Find $\partial f(x)$ for $x > 0$ and for $x < 0$.
- (c) For each $g \in \partial f(0)$, write the corresponding global affine lower bound.

Subgradients of the absolute value function

At 0, a number g is a subgradient exactly when

$$|y| \geq gy \quad \text{for every } y \in \mathbb{R}.$$

Positive y requires $g \leq 1$, while negative y requires $g \geq -1$. Hence

$$\partial f(0) = [-1, 1], \quad \partial f(x) = \begin{cases} \{1\}, & x > 0, \\ \{-1\}, & x < 0. \end{cases}$$

Each $g \in [-1, 1]$ gives the global lower bound $|y| \geq gy$.

Section 4

Second-order characterizations

Second-order test in 1D

Let $S \subseteq \mathbb{R}$ be an interval and let f be twice differentiable on an open interval containing S :

Weak inequalities.

$$f'' \geq 0 \text{ on } S \Leftrightarrow f \text{ convex on } S,$$

$$f'' \leq 0 \text{ on } S \Leftrightarrow f \text{ concave on } S.$$

Strict inequalities.

$$f'' > 0 \text{ on } S \Rightarrow f \text{ strictly convex on } S,$$

$$f'' < 0 \text{ on } S \Rightarrow f \text{ strictly concave on } S.$$

Note the directions of implications. The **weak** inequalities **characterize** convexity and concavity. The **strict** inequalities are **sufficient** but **not necessary** conditions.

Video: Joydeep Dutta, *Twice differentiable convex functions*.

Strict convexity with zero curvature at one point

Verify the definition. Fix $x \neq y$ and $\lambda \in (0, 1)$, and set $m = \lambda x + (1 - \lambda)y$. Then

$$\lambda x^2 + (1 - \lambda)y^2 - m^2 = \lambda(1 - \lambda)(x - y)^2 > 0.$$

Hence

$$m^4 < (\lambda x^2 + (1 - \lambda)y^2)^2 \leq \lambda x^4 + (1 - \lambda)y^4,$$

where the second inequality follows from

$$\lambda(1 - \lambda)(x^2 - y^2)^2 \geq 0.$$

Thus $f(m) < \lambda f(x) + (1 - \lambda)f(y)$: $f(x) = x^4$ is strictly convex by definition.

THE CURVATURE TEST IS ONE-WAY

$$f(x) = x^4, \quad f'(x) = 4x^3, \quad f''(x) = 12x^2.$$

At the origin,

$$f''(0) = 0.$$

$f'' > 0$ everywhere $\Rightarrow$ strict convexity,
but the reverse implication fails.

A strictly concave quadratic

Let

$$f(x) = 10x - x^2, \quad x \geq 0.$$

The first-order condition is

$$f'(x) = 10 - 2x = 0 \quad \Rightarrow \quad x^* = 5.$$

Since

$$f''(x) = -2 < 0,$$

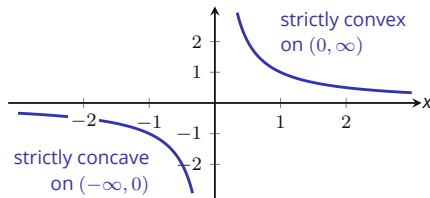
the second-order criterion implies that f is strictly concave on the convex feasible set $\mathbb{R}_+$. Therefore the stationary point $x^* = 5$ is the unique global maximizer.

$$f(5) = 10 \cdot 5 - 5^2 = 25.$$

The domain is part of the statement

CURVATURE CAN CHANGE ACROSS REGIONS

For $f(x) = x^3$ we get $f'(x) = 6x$, so f is strictly convex on $(0, \infty)$, strictly concave on $(-\infty, 0)$, and neither on $\mathbb{R}$, where f' changes sign.



THE NATURAL DOMAIN CAN FAIL TO BE CONVEX

For $f(x) = 1/x$ we get $f'(x) = 2/x^3$, positive on $(0, \infty)$ and negative on $(-\infty, 0)$. Its natural domain skips the midpoint: $-1, 1 \in \mathbb{R} \setminus \{0\}$ while $\frac{-1+1}{2} = 0 \notin \mathbb{R} \setminus \{0\}$.

Domain rule. Convexity and concavity are stated on a convex domain. For $f(x) = 1/x$, use the two separate intervals: strict concavity on $(-\infty, 0)$ and strict convexity on $(0, \infty)$.

From Taylor curvature to a matrix sign test

In one variable the criterion is the sign of the number $f''(x)$. In $\mathbb{R}^n$ the second derivative is the Hessian matrix $H_f(x)$, and the second-order expansion from Differentiation says what its sign has to mean:

$$f(x+h) - f(x) = \nabla f(x)^\top h + \frac{1}{2} h^\top H_f(x) h + o(\|h\|^2).$$

The second-order term is

$$Q_x(h) := h^\top H_f(x) h,$$

a quadratic form whose representing matrix is the Hessian.

So the n -dimensional analogue of $f''(x) \geq 0$ is a statement about the sign of $Q_x(h)$ across directions h , which is what the matrix tests below decide.

Video: [Ketan Rajawat, First and second order conditions for convexity.](#)

Hessian signs across a domain

This section uses the Hessian to classify curvature *across a domain*.

Hessian criterion on a domain. If $Q_x(h) = h^\top H_f(x)h \geq 0$ for every direction h and every point x of a convex domain, the matrix test characterizes convexity on that domain; the reversed inequality characterizes concavity. The next pages name these sign patterns and give ways to check them.

At one critical point. If $\nabla f(x^*) = 0$, the sign of $Q_{x^*}(h)$ in nearby directions is the second-order test for a local maximum, minimum, or saddle point. That sign at the single point x^* does not settle convexity or concavity on the domain.

Definiteness

For a twice differentiable function, second-order change in a direction h is governed by $h^\top H_f(x)h$. A **quadratic form** on $\mathbb{R}^n$ can be written $Q(x) = x^\top Ax$; take $A = A^\top$ because only the symmetric part affects $x^\top Ax$.

Positive definite

$$x^\top Ax > 0 \text{ for all } x \neq 0.$$

Negative definite

$$x^\top Ax < 0 \text{ for all } x \neq 0.$$

Positive semidefinite

$$x^\top Ax \geq 0 \text{ for all } x.$$

Negative semidefinite

$$x^\top Ax \leq 0 \text{ for all } x.$$

Indefinite

$$x^\top Ax > 0 > y^\top Ay \text{ for some } x, y.$$

An indefinite form is positive in some directions and negative in others.

Covariance matrices and portfolio variance

Every covariance matrix is positive semidefinite. Let $\Sigma = \text{Cov}(X)$. For every $a \in \mathbb{R}^n$,

$$a^\top \Sigma a = \text{Var}(a^\top X) \geq 0.$$

Hence $\Sigma \succeq 0$. It is positive definite exactly when

$$\text{Var}(a^\top X) > 0 \quad \text{for every } a \neq 0.$$

A DEPENDENT PAIR If $X = (Z, 2Z)^\top$, then

$$\Sigma = \text{Var}(Z) \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

For $a = (2, -1)^\top \neq 0$,

$$a^\top X = 0, \quad a^\top \Sigma a = 0.$$

This nonzero linear combination has zero variance, so $\Sigma \succeq 0$ but $\Sigma \not\succ 0$.

Video: [Brunei Math Club, Why every covariance matrix is positive semidefinite.](#)

Homogeneity and sign along rays

Scaling a quadratic form

For $Q(x) = x^T A x$ and any scalar $\lambda \in \mathbb{R}$,

$$Q(\lambda x) = (\lambda x)^T A (\lambda x) = \lambda^2 x^T A x = \lambda^2 Q(x).$$

Thus Q is homogeneous of degree two; in particular, $Q(0) = 0$ and $Q(-x) = Q(x)$. If $x \neq 0$ and $\lambda \neq 0$, then $\lambda^2 > 0$, so

$$\text{sign } Q(\lambda x) = \text{sign } Q(x).$$

On each line through the origin, Q is positive away from the origin, negative away from the origin, or identically zero.

Further reading: [Boyd and Vandenberghe, quadratic-form slides](#).

Geometry: an indefinite quadratic form

For

$$Q(x_1, x_2) = x_1(x_1 - 4x_2),$$

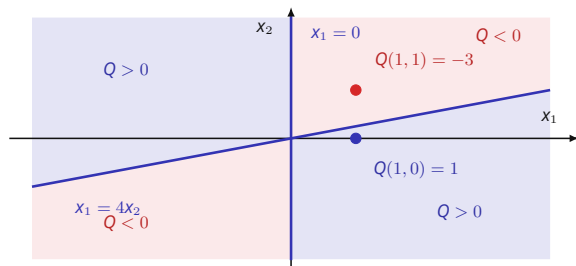
the zero directions are $x_1 = 0$ and $x_1 = 4x_2$.

Moreover,

$$Q(1, 0) = 1 > 0, \quad Q(1, 1) = -3 < 0.$$

The zero lines separate positive and negative directional sectors, so Q is indefinite.

Definiteness concerns the sign across all directions, not the value at one vector.



Each open sector has a constant sign.

Examples of definiteness

POSITIVE DEFINITE

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad x^\top A x = 2x_1^2 + x_2^2 > 0$$

for every $x \neq 0$.

POSITIVE SEMIDEFINITE

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad x^\top B x = (x_1 + x_2)^2 \geq 0.$$

But for $x = (1, -1)^\top$, $x^\top B x = 0$.

NEGATIVE DEFINITE

$$C = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad x^\top C x = -x_1^2 - 2x_2^2 < 0$$

for every $x \neq 0$.

INDEFINITE

$$D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad x^\top D x = x_1^2 - x_2^2.$$

It is positive at $(1, 0)^\top$ and negative at $(0, 1)^\top$.

Definiteness and eigenvalues

Let $A = A^T$. Its eigenvalues are real, and their signs give an equivalent definiteness test:

Eigenvalue sign test.

A positive definite	$\Leftrightarrow$	$\lambda_i(A) > 0$ for every i ,
A positive semidefinite	$\Leftrightarrow$	$\lambda_i(A) \geq 0$ for every i ,
A negative definite	$\Leftrightarrow$	$\lambda_i(A) < 0$ for every i ,
A negative semidefinite	$\Leftrightarrow$	$\lambda_i(A) \leq 0$ for every i .

A symmetric matrix is indefinite exactly when it has at least one positive and at least one negative eigenvalue. Eigenvalues and principal-minor tests are alternative ways to determine the same sign structure.

Further reading: [Boyd and Vandenberghe, quadratic-form slides](#).

Principal submatrices and principal minors

Let A be an $n \times n$ matrix and let $I \subseteq \{1, \dots, n\}$ contain k indices.

Principal submatrix of order k . $A[I, I]$

keeps the rows indexed by I and the same columns indexed by I .

Principal minor of order k .

$M_I := \det A[I, I]$, one for each of the $\binom{n}{k}$ index sets.

A **leading** principal submatrix uses

$I = \{1, \dots, k\}$, with determinant

$D_k = \det A[I, I]$.

WORKED INDEX SET: $I = \{1, 3\}$

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 5 \end{bmatrix} \Rightarrow A[I, I] = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix},$$

so $M_{\{1,3\}} = 20 - 4 = 16$: row 2 and column 2 are removed together.

For $n = 3$ the index sets are $\{1\}, \{2\}, \{3\}; \{1, 2\}, \{1, 3\}, \{2, 3\}; \{1, 2, 3\}$: seven principal minors, of which D_1, D_2, D_3 are leading. An $n \times n$ matrix has $2^n - 1$ principal minors and n leading ones.

Video: [The Complete Guide, Sylvester's criterion](#), on the leading minors D_k .

Definiteness via minors

Let $A = A^\top$, let D_k be its leading principal minors, and let $M_I = \det A[I, I]$.

Definite: n leading minors.

$$A \succ 0 \Leftrightarrow D_k > 0 \quad (k = 1, \dots, n),$$

$$A \prec 0 \Leftrightarrow (-1)^k D_k > 0 \quad (k = 1, \dots, n).$$

Semidefinite: all principal minors.

$$A \succeq 0 \Leftrightarrow M_I \geq 0 \quad (I \neq \emptyset),$$

$$A \preceq 0 \Leftrightarrow (-1)^{|I|} M_I \geq 0 \quad (I \neq \emptyset).$$

Strict sign + leading minors

tests definiteness;

weak sign + every principal minor

tests semidefiniteness.

For negative semidefiniteness, odd-order principal minors are nonpositive, even-order principal minors are nonnegative, and zeros are allowed.

Leading minors do not test semidefiniteness

When zeros appear in the leading sequence, a wrongly signed principal minor can sit on an omitted index set.

LEADING MINORS LOOK ADMISSIBLE

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix},$$

$$D_1 = 1 > 0, \quad D_2 = 0, \quad D_3 = \det A = 0,$$

so every leading principal minor is nonnegative.

The omitted index set $I = \{1, 3\}$.

$$A[I, I] = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}, \quad M_{\{1,3\}} = 2 - 4 = -2 < 0.$$

Positive semidefiniteness requires *every* principal minor to be nonnegative, so this one minor rules it out.

THE SAME FAILURE IN $x^T A x$

With $Q(x) = x_1^2 + 4x_1x_3 + 2x_3^2$, $Q(1, 0, 1) = 7 > 0$ and $Q(1, 0, -1) = -1 < 0$, so A is indefinite: the omitted minor and the two directions detect one sign failure.

Video: [Arizona Math Camp, Lecture 29\(C\): quadratic forms: determinant conditions for definiteness.](#)

The 2×2 case

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix},$$

$$\det A = a_{11}a_{22} - a_{12}^2.$$

$$\det A < 0 \quad A \text{ indefinite}$$

$$\det A > 0, a_{11} > 0 \quad A \succ 0$$

$$\det A > 0, a_{11} < 0 \quad A \prec 0$$

$$\det A \geq 0, a_{11} \geq 0, a_{22} \geq 0 \quad A \succeq 0$$

$$\det A \geq 0, a_{11} \leq 0, a_{22} \leq 0 \quad A \preceq 0$$

The definite rows need only a_{11} , because $\det A > 0$ forces $a_{11}a_{22} > a_{12}^2 \geq 0$, so the two diagonal entries carry the same strict sign. That implication disappears at $\det A = 0$, so the semidefinite rows check both diagonal entries, which are the principal minors of order one.

The 3×3 case

For a symmetric matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix},$$

the leading principal minors are

$$D_1 = a_{11},$$

$$D_2 = \det \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}, \quad D_3 = \det A.$$

Hence Sylvester's criterion specializes to

$$A \succ 0 \Leftrightarrow D_1 > 0, D_2 > 0, D_3 > 0,$$

$$A \prec 0 \Leftrightarrow D_1 < 0, D_2 > 0, D_3 < 0.$$

Definiteness also fixes the sign of the other four principal minors: if $A \succ 0$ all seven are positive, and if $A \prec 0$ each minor of order k has sign $(-1)^k$. Testing A still takes only the three leading minors D_1, D_2, D_3 ; the remaining four follow.

Portfolio risk matrix

From $\Sigma \succeq 0$ to $\Sigma \succ 0$ for one covariance matrix

Variance already gives $\Sigma \succeq 0$ for every covariance matrix; the minors settle the strict case for a given one. For three asset returns with

$$\Sigma = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad w^\top \Sigma w = \text{Var}(w^\top X) :$$

LEADING PRINCIPAL MINORS

$$D_1 = 4, \quad D_2 = \begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} = 11,$$

$$D_3 = \det \Sigma = 18.$$

All positive, so $\Sigma \succ 0$: every portfolio with $w \neq 0$ carries strictly positive variance.

THE OTHER FOUR PRINCIPAL MINORS

$$\text{Order 1: } 4, 3, 2; \quad \text{order 2: } \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = 5, \quad \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} = 8.$$

Positive, as $\Sigma \succ 0$ requires, and none of them entered the test.

Workflow for definiteness tests

Let $A = A^\top$. The minors route proceeds in three steps.

1. LEADING MINORS

Match

$+, +, \dots, +$

or

$-, +, -, \dots$

If one matches, A is definite.



2. ALL PRINCIPAL MINORS

If Step 1 fails, test

$$M_l \geq 0 \quad \forall l$$

and

$$(-1)^{|l|} M_l \geq 0 \quad \forall l.$$



3. CLASSIFY A weak sign pattern gives $A \succeq 0$ or $A \preceq 0$.

If neither holds, A is indefinite. Two vectors giving opposite signs certify this directly.

Eigenvalues provide a one-pass alternative: their signs classify every symmetric matrix directly. The Hessian criterion on the next page turns this matrix classification into a convexity or concavity classification.

Definiteness from principal minors

Classify each symmetric matrix using principal minors:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

For each matrix, state whether the leading-minor test settles the answer. If it does not, inspect all principal minors or exhibit vectors with quadratic forms of opposite signs.

Definiteness from principal minors

MATRIX A

$$D_1 = 2 > 0, \quad D_2 = 3 > 0.$$

Thus $A \succ 0$.

MATRIX B Its leading determinant is zero. All principal minors are 1, 1, 0, so $B \succeq 0$, but $B \not\succ 0$.

MATRIX C $D_2 = 1 - 4 = -3 < 0$. Also $e_1^T C e_1 = 1 > 0$, while $(1, -1)C(1, -1)^T = -2 < 0$. Thus C is indefinite.

Second-order test in $\mathbb{R}^n$

Let $S \subseteq \mathbb{R}^n$ be open and convex, and let $f: S \rightarrow \mathbb{R}$ be twice continuously differentiable.

NECESSARY AND SUFFICIENT $\Leftrightarrow$

$$\begin{aligned} f \text{ convex on } S &\Leftrightarrow H_f(x) \succeq 0 \quad \text{for every } x \in S, \\ f \text{ concave on } S &\Leftrightarrow H_f(x) \preceq 0 \quad \text{for every } x \in S. \end{aligned}$$

SUFFICIENT ONLY $\Rightarrow$

$$\begin{aligned} H_f(x) \succ 0 \text{ for every } x \in S &\Rightarrow f \text{ strictly convex on } S, \\ H_f(x) \prec 0 \text{ for every } x \in S &\Rightarrow f \text{ strictly concave on } S. \end{aligned}$$

Hessian test

STRICTLY CONVEX Let $f(x, y) = x^2 + xy + y^2$. Then

$$H_f = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Leading minors are

$$D_1 = 2 > 0, \quad D_2 = 4 - 1 = 3 > 0.$$

Hence H_f is positive definite.

Video: [MIT OpenCourseWare, Positive Definite Matrices and Minima.](#)

STRICTLY CONCAVE Let $g(x, y) = \ln x + \ln y$ on $\mathbb{R}_{++}^2$. Then

$$H_g = \begin{bmatrix} -1/x^2 & 0 \\ 0 & -1/y^2 \end{bmatrix}.$$

For $v \neq 0$,

$$v^\top H_g v = -\frac{v_1^2}{x^2} - \frac{v_2^2}{y^2} < 0.$$

Classifying functions with Hessians

Compute the gradient and Hessian, then classify each function on its full Euclidean domain:

$$f(x, y) = x^3 + y^2, \quad g(x, y, z) = 3x^2 + y^2 - 2xy + e^z.$$

Classifying functions with Hessians

GRADIENTS, HESSIANS, AND CLASSIFICATIONS

For the first function,

$$\nabla f(x, y) = \begin{bmatrix} 3x^2 \\ 2y \end{bmatrix},$$

$$H_f(x, y) = \begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}.$$

Its sign changes with x , so f is neither convex nor concave on $\mathbb{R}^2$.

For the second,

$$\nabla g(x, y, z) = \begin{bmatrix} 6x - 2y \\ 2y - 2x \\ e^z \end{bmatrix},$$

$$H_g(x, y, z) = \begin{bmatrix} 6 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & e^z \end{bmatrix}.$$

Its leading principal minors are 6, 8, and $8e^z$, all positive. Thus H_g is positive definite and g is strictly convex on $\mathbb{R}^3$.

One property, several equivalent views

Let $S \subseteq \mathbb{R}^n$ be open and convex and let $f: S \rightarrow \mathbb{R}$ be twice continuously differentiable.

Equivalent statements for convexity. Each statement is equivalent to f being convex:

- ◆ f satisfies the chord inequality.
- ◆ $\text{epi } f$ is convex.
- ◆ $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$.
- ◆ $\partial f(x) \neq \emptyset$ for every $x \in S$.
- ◆ $H_f(x) \succeq 0$ for every $x \in S$.

Equivalent statements for concavity. Each statement is equivalent to f being concave:

- ◆ f satisfies the reversed chord inequality.
- ◆ $\text{hypo } f$ is convex.
- ◆ $f(y) \leq f(x) + \nabla f(x)^\top (y - x)$.
- ◆ $\partial^+ f(x) \neq \emptyset$ for every $x \in S$.
- ◆ $H_f(x) \preceq 0$ for every $x \in S$.

Each list characterizes the weak property. Definite Hessians give one-way tests for the corresponding strict property.

Notation in the second and fourth bullets

$$\begin{aligned} \text{epi } f &= \{(x, t) : t \geq f(x)\}, \quad \text{hypo } f = \{(x, t) : t \leq f(x)\}, \quad \partial f(x) = \{g : f(y) \geq f(x) + g^\top (y - x) \forall y\}, \\ \partial^+ f(x) &= \{g : f(y) \leq f(x) + g^\top (y - x) \forall y\}. \end{aligned}$$

Section 5

Quasiconvexity & quasiconcavity

Definitions & contour-set link

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$. For a level $c \in \mathbb{R}$,

$$U_f(c) = \{x \in S : f(x) \geq c\}, \quad L_f(c) = \{x \in S : f(x) \leq c\},$$

the upper and lower contour sets. The level set from the Functions section is their intersection, $\{x : f(x) = c\} = U_f(c) \cap L_f(c)$.

Quasiconvex. $f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}$ for all $x, y \in S$ and $\lambda \in [0, 1]$; equivalently, every lower contour set $L_f(c)$ is convex.

Quasiconcave. $f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$ for all $x, y \in S$ and $\lambda \in [0, 1]$; equivalently, every upper contour set $U_f(c)$ is convex.

Videos: [EZ Economics, Understanding quasiconcave and quasiconvex functions](#); [Arizona Math Camp, Upper and lower contour sets and quasiconcavity](#) (from 16:55).

Convexity versus quasiconvexity

f convex $\Rightarrow f$ quasiconvex $\Leftrightarrow L_f(c)$ convex for every c ,

f concave $\Rightarrow f$ quasiconcave $\Leftrightarrow U_f(c)$ convex for every c .

The first arrow runs one way; the second is an equivalence.

THE IMPLICATION IN ONE LINE

If f is convex, then

$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$,
and a weighted average is at most the maximum, so the right side is at most $\max\{f(x), f(y)\}$. Reversing the inequalities gives the concave case with min.

WHAT EACH SIDE ASKS OF THE GRAPH

Convexity constrains the whole epigraph in $\mathbb{R}^{n+1}$: f convex $\Leftrightarrow \text{epi } f$ convex, and f concave $\Leftrightarrow \text{hypo } f$ convex. Quasiconvexity asks only that each horizontal slice $L_f(c)$ be convex, so it can hold where convexity fails.

Quasiconcave but not concave

Let $u(x, y) = xy$ on $\mathbb{R}_{++}^2$. If $c \leq 0$, then $U_u(c) = \mathbb{R}_{++}^2$, which is convex. If $c > 0$,

$$U_u(c) = \{(x, y) : xy \geq c\} = \{(x, y) : y \geq c/x, x > 0\}.$$

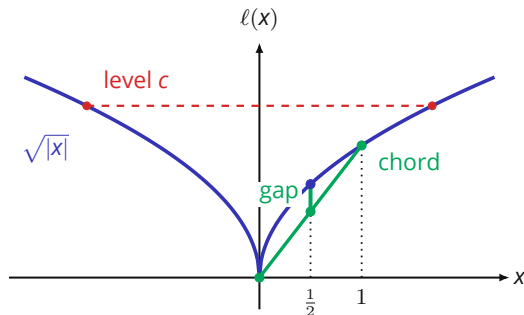
Since $d^2(c/x)/dx^2 = 2c/x^3 > 0$ for $x > 0$, the function c/x is convex, so its epigraph is convex. Hence every upper contour set of u is convex, and therefore u is quasiconcave.

$$H_u = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The Hessian is indefinite, so u is not concave.

Quasiconvex but not convex

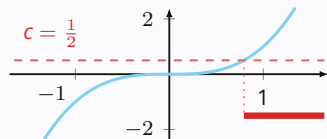
A forecaster incurs a loss based on the signed forecast error $x \in \mathbb{R}$: $\ell(x) = \sqrt{|x|}$. The loss increases with the absolute error, but at a decreasing rate. For $c \geq 0$, its sublevel sets are intervals, $\{x : \ell(x) \leq c\} = [-c^2, c^2]$; for $c < 0$, they are empty. Hence ℓ is quasiconvex. The chord joining $(0, \ell(0))$ and $(1, \ell(1))$ settles the other half: at its midpoint $\ell(1/2) = 1/\sqrt{2} > 1/2 = [\ell(0) + \ell(1)]/2$, so the graph rises above the chord and convexity fails.



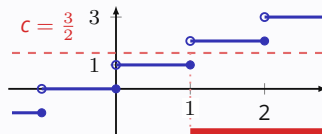
Red level line: the sublevel set is an interval. Green chord:
at $x = \frac{1}{2}$ the graph sits $\frac{1}{\sqrt{2}} - \frac{1}{2}$ above it.

Three functions that test the definition

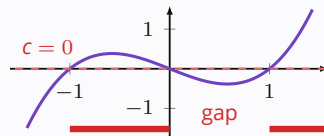
Figure 5 · The bar under each axis is the upper contour set at the level drawn.



$f(x) = x^3$. Each upper contour set is a half-line, so f is quasiconcave, and so is $-f$. Yet $f''(x) = 6x$ changes sign, so f is neither concave nor convex.



$f(x) = \lceil x \rceil$. The definition asks nothing about continuity: this step function is quasiconcave and quasiconvex while jumping at every integer.



$f(x) = x^3 - x$. Now $U_f(0) = [-1, 0] \cup [1, \infty)$ falls into two pieces, so f is not quasiconcave, though it adds the quasiconcave x^3 to the linear $-x$.

The step function is Example 3.29 of [Boyd and Vandenberghe](#). The right panel is where quasiconcavity parts company with concavity: a sum of concave functions stays concave, a sum of quasiconcave functions need not stay quasiconcave.

More examples: [EconJohn, Quasiconcave and quasiconvex functions: an introduction](#).

Convex preferences

Let $u : S \rightarrow \mathbb{R}$ represent a preference relation $\succeq$ on a convex set S . Preferences are **convex** when

$$x \succeq z, y \succeq z \Rightarrow \lambda x + (1 - \lambda)y \succeq z \quad \forall \lambda \in [0, 1].$$

Since $\{x : x \succeq z\} = \{x : u(x) \geq u(z)\}$,

$$\boxed{\succeq \text{ is convex} \Leftrightarrow u \text{ is quasiconcave.}}$$

Strict quasiconcavity requires, for $x \neq y$ and $\lambda \in (0, 1)$,

$$u(\lambda x + (1 - \lambda)y) > \min\{u(x), u(y)\}.$$

Under the corresponding strict-convexity definition for preferences, the strict versions are equivalent as well.

Preference convexity is a property of the upper contour sets. Concavity of a utility representation is stronger: it implies quasiconcavity, while an increasing transformation can change concavity.

Video: [Jochumzen, Convex preferences.](#)

Ordinal transformations

Compare

$$u(x, y) = xy, \quad v(x, y) = \ln x + \ln y = \ln u(x, y).$$

Since $\ln$ is strictly increasing,

$$u(x, y) \geq u(x', y') \Leftrightarrow v(x, y) \geq v(x', y').$$

Thus u and v represent the same preferences.

Yet v is concave and u is not:

$H_v = \text{diag}(-1/x^2, -1/y^2)$ is negative definite, while $H_u = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues ± 1 .

Here $u = \exp(v)$ with $\exp$ increasing: an increasing transformation of a concave function need not be concave. Quasiconcavity does survive.

Quick check. If u is quasiconcave and ϕ is strictly increasing, then $\phi \circ u$ is quasiconcave. Concavity need not be preserved.

Quasiconvexity and quasiconcavity

For each function below, determine whether it is quasiconvex, quasiconcave, both, or neither on $\mathbb{R}$:

$$f(x) = e^x, \quad g(x) = x^2, \quad h(x) = -|x|.$$

Justify every answer using upper or lower contour sets. Do not use a Hessian test.

Quasiconvexity and quasiconcavity

e^x is strictly increasing. Its upper and lower contour sets are half-lines, so it is both quasiconvex and quasiconcave.

The lower contour sets of x^2 are intervals $[-\sqrt{c}, \sqrt{c}]$ for $c \geq 0$, so x^2 is quasiconvex. For $c > 0$, its upper contour set is the union of two rays, so it is not quasiconcave.

The upper contour sets of $-|x|$ are intervals $[-c, c]$ after writing the nonempty levels as $c \geq 0$ for $-|x| \geq -c$. Hence $-|x|$ is quasiconcave. Its lower contour sets can be two rays, so it is not quasiconvex.

What it gives in maximization

Let $S \subseteq \mathbb{R}^n$ be convex and let f be quasiconcave on S .

Convex upper contour sets

If $f(x) \geq c$ and $f(y) \geq c$, then

$$f(\lambda x + (1 - \lambda)y) \geq c \quad (0 \leq \lambda \leq 1).$$

Thus mixtures of choices that attain at least c also attain at least c .

A first-order implication

If f is differentiable near S , then

$$f(y) \geq f(x) \Rightarrow \nabla f(x)^\top (y - x) \geq 0.$$

If $\nabla f(x) \neq 0$, the equality hyperplane supports the upper contour set at x .

Uniqueness. If f is strictly quasiconcave, it has at most one maximizer on S . Quasiconcavity alone does not make stationarity sufficient: $f(x) = x^3$ is strictly quasiconcave on $[-1, 1]$, but $f'(0) = 0$ and the maximizer is 1.

Further reading: [Walker, Differentiable Quasiconcave Functions](#).